

Report: User & Group Management

1. Introduction

Effective user and group management is a critical component of maintaining a secure and well-organized Active Directory (AD) environment. This set of tasks focused on establishing essential user identities and departmental group structures within a Windows Server domain. Using both graphical tools and PowerShell automation, we created user accounts, enforced password and security settings, and structured access through logical security groups. The operations performed laid the groundwork for scalable role-based access control and policy application across the domain. All actions were executed on the domain controller and validated through system tools to ensure proper application and visibility.

2. Tools used

- Windows Server (AD DS)
- Active Directory Users and Computers (ADUC)
- Windows PowerShell (ActiveDirectory Module)
- Group Policy Management Console (GPMC)

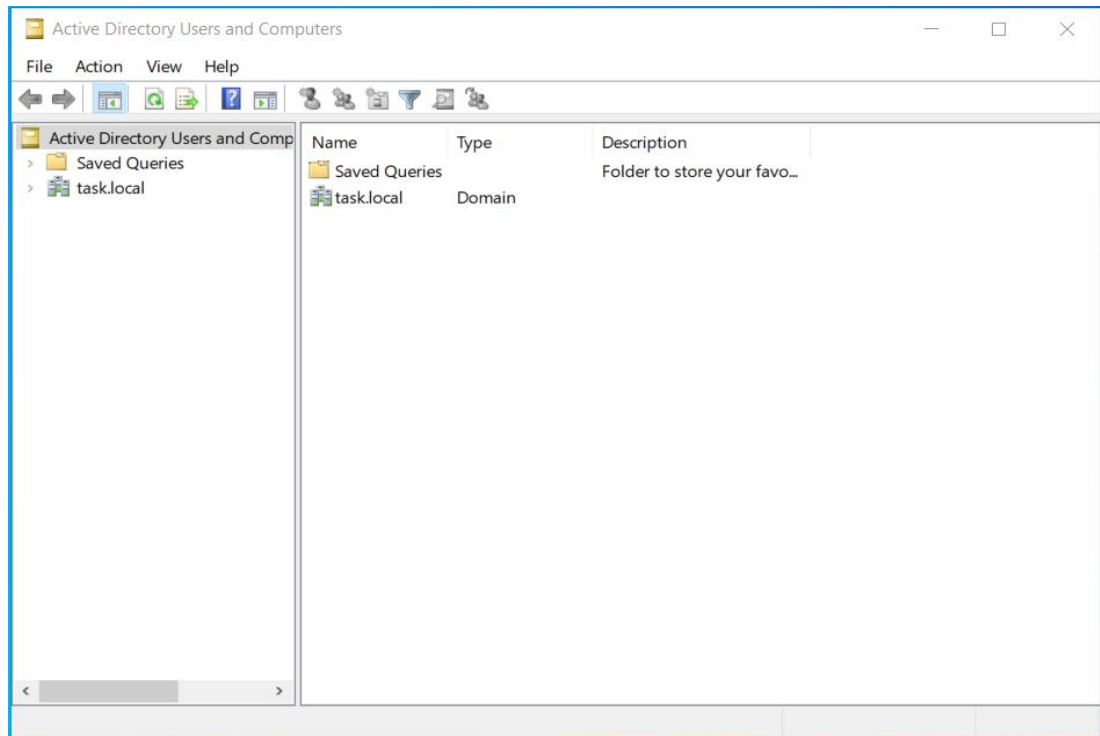
3. Objective

The objective of this task set was to implement foundational Active Directory user and group management practices within a domain environment. This included creating user accounts aligned to departmental structures, resetting and enforcing password policies, disabling inactive accounts, organizing users into security groups, and verifying group memberships. These operations aimed to establish a secure and well-structured identity framework that supports access control, and future policy enforcement. Each action was executed using standard PowerShell cmdlets and validated through both ADUC and command-line output to ensure accuracy and compliance with domain-level administrative standards.

4. Methodology

4.1 Create 10 User Accounts

PowerShell was used on the Windows Server to create 10 user accounts with names in the First.Department format. Each account was enabled at creation and assigned a default password using ConvertTo-SecureString. The accounts were verified in ADUC under the Users container.



(NO AD users at first)

```
Administrator: Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> New-ADUser -Name "Aarav.Finance" -GivenName "Aarav" -Surname "Finance" -SamAccountName "Aarav.Finance" -UserPrincipalName "Aarav.Finance@yourdomain.local" -AccountPassword (ConvertTo-SecureString "Password@123" -AsPlainText -Force) -Enabled $true

PS C:\Users\Administrator> New-ADUser -Name "Diya.Finance" -GivenName "Diya" -Surname "Finance" -SamAccountName "Diya.Finance" -UserPrincipalName "Diya.Finance@yourdomain.local" -AccountPassword (ConvertTo-SecureString "Password@123" -AsPlainText -Force) -Enabled $true

PS C:\Users\Administrator> New-ADUser -Name "Kabir.IT" -GivenName "Kabir" -Surname "IT" -SamAccountName "Kabir.IT" -UserPrincipalName "Kabir.IT@yourdomain.local" -AccountPassword (ConvertTo-SecureString "Password@123" -AsPlainText -Force) -Enabled $true

PS C:\Users\Administrator> New-ADUser -Name "Meera.Finance" -GivenName "Meera" -Surname "Finance" -SamAccountName "Meera.Finance" -UserPrincipalName "Meera.Finance@yourdomain.local" -AccountPassword (ConvertTo-SecureString "Password@123" -AsPlainText -Force) -Enabled $true

PS C:\Users\Administrator> New-ADUser -Name "Rohan.HR" -GivenName "Rohan" -Surname "HR" -SamAccountName "Rohan.HR" -UserPrincipalName "Rohan.HR@yourdomain.local" -AccountPassword (ConvertTo-SecureString "Password@123" -AsPlainText -Force) -Enabled $true

PS C:\Users\Administrator> New-ADUser -Name "Ananya.HR" -GivenName "Ananya" -Surname "HR" -SamAccountName "Ananya.HR" -UserPrincipalName "Ananya.HR@yourdomain.local" -AccountPassword (ConvertTo-SecureString "Password@123" -AsPlainText -Force) -Enabled $true

PS C:\Users\Administrator> New-ADUser -Name "Yash.IT" -GivenName "Yash" -Surname "IT" -SamAccountName "Yash.IT" -UserPrincipalName "Yash.IT@yourdomain.local" -AccountPassword (ConvertTo-SecureString "Password@123" -AsPlainText -Force) -Enabled $true

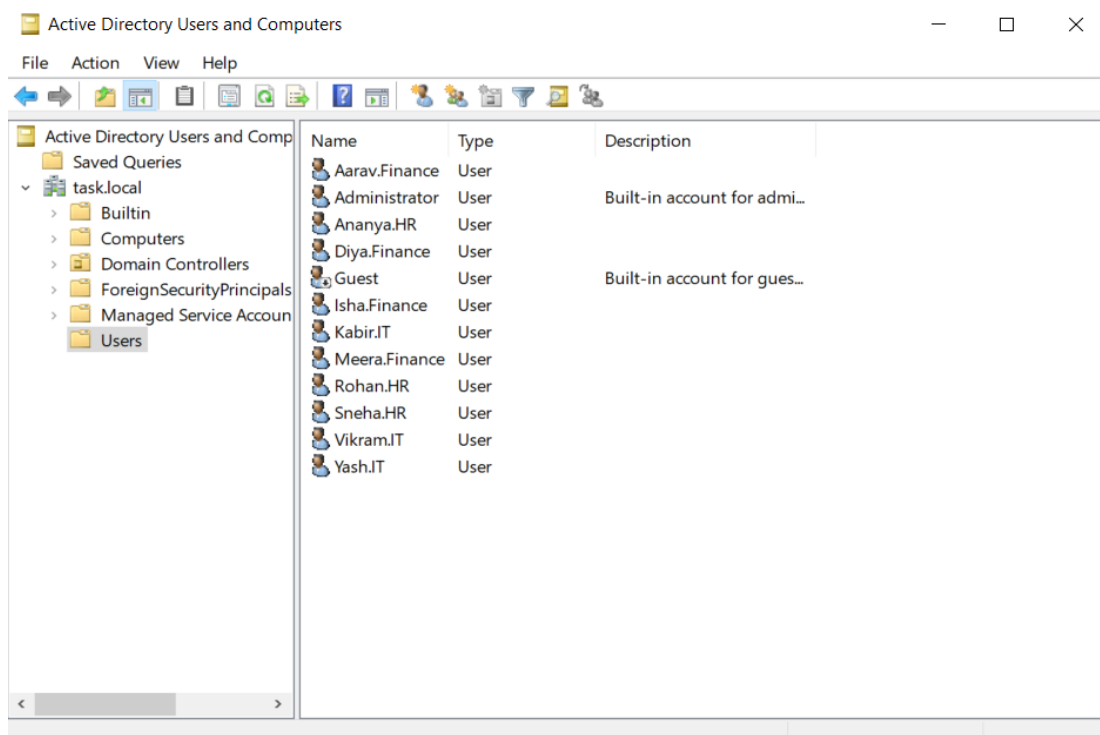
PS C:\Users\Administrator> New-ADUser -Name "Isha.Finance" -GivenName "Isha" -Surname "Finance" -SamAccountName "Isha.Finance" -UserPrincipalName "Isha.Finance@yourdomain.local" -AccountPassword (ConvertTo-SecureString "Password@123" -AsPlainText -Force) -Enabled $true

PS C:\Users\Administrator> New-ADUser -Name "Vikram.IT" -GivenName "Vikram" -Surname "IT" -SamAccountName "Vikram.IT" -UserPrincipalName "Vikram.IT@yourdomain.local" -AccountPassword (ConvertTo-SecureString "Password@123" -AsPlainText -Force) -Enabled $true

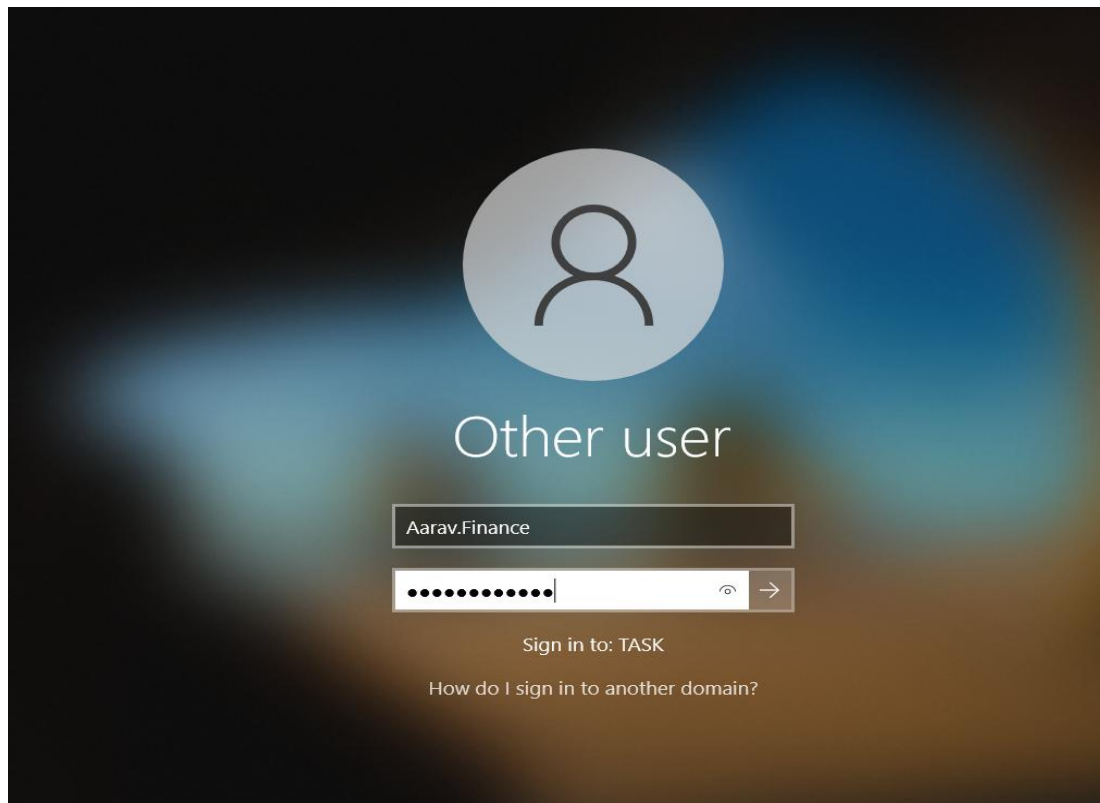
PS C:\Users\Administrator> New-ADUser -Name "Sneha.HR" -GivenName "Sneha" -Surname "HR" -SamAccountName "Sneha.HR" -UserPrincipalName "Sneha.HR@yourdomain.local" -AccountPassword (ConvertTo-SecureString "Password@123" -AsPlainText -Force) -Enabled $true
```

(We ran commands in power shell to add 10 users)

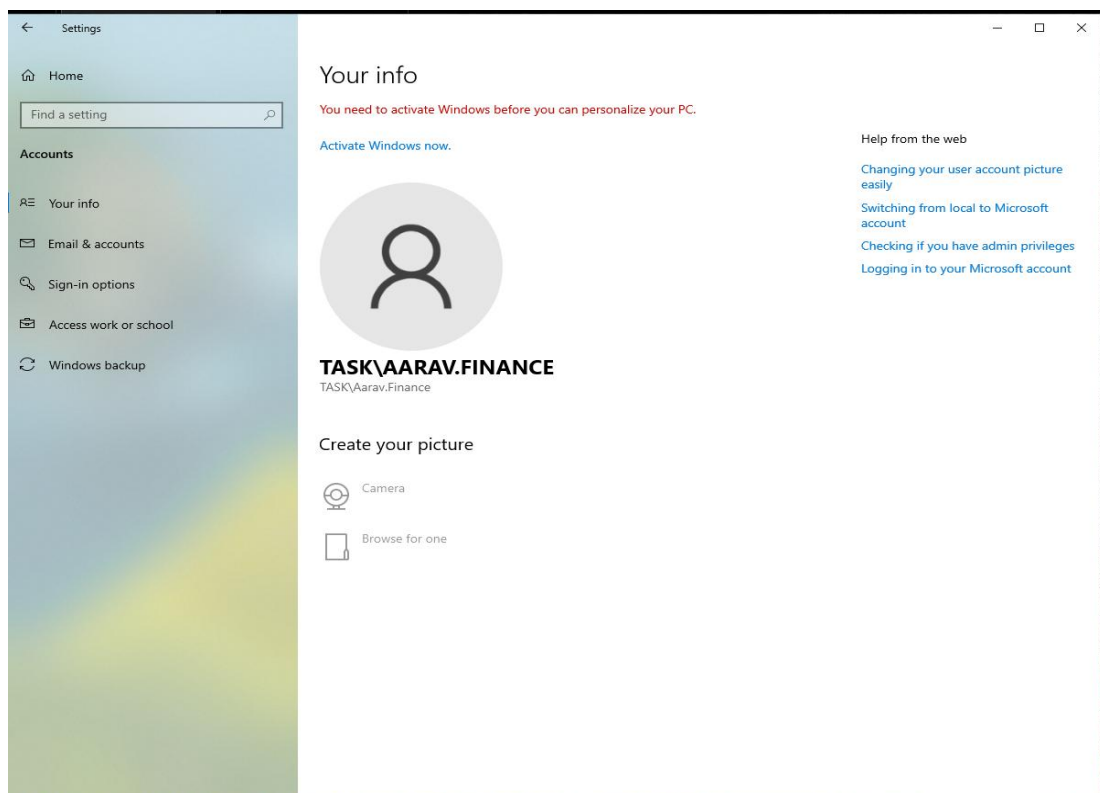
(Password@123)



(Successfully created users)



(To verify- sign in with new user)



(successfully signed-in as the new user we created)

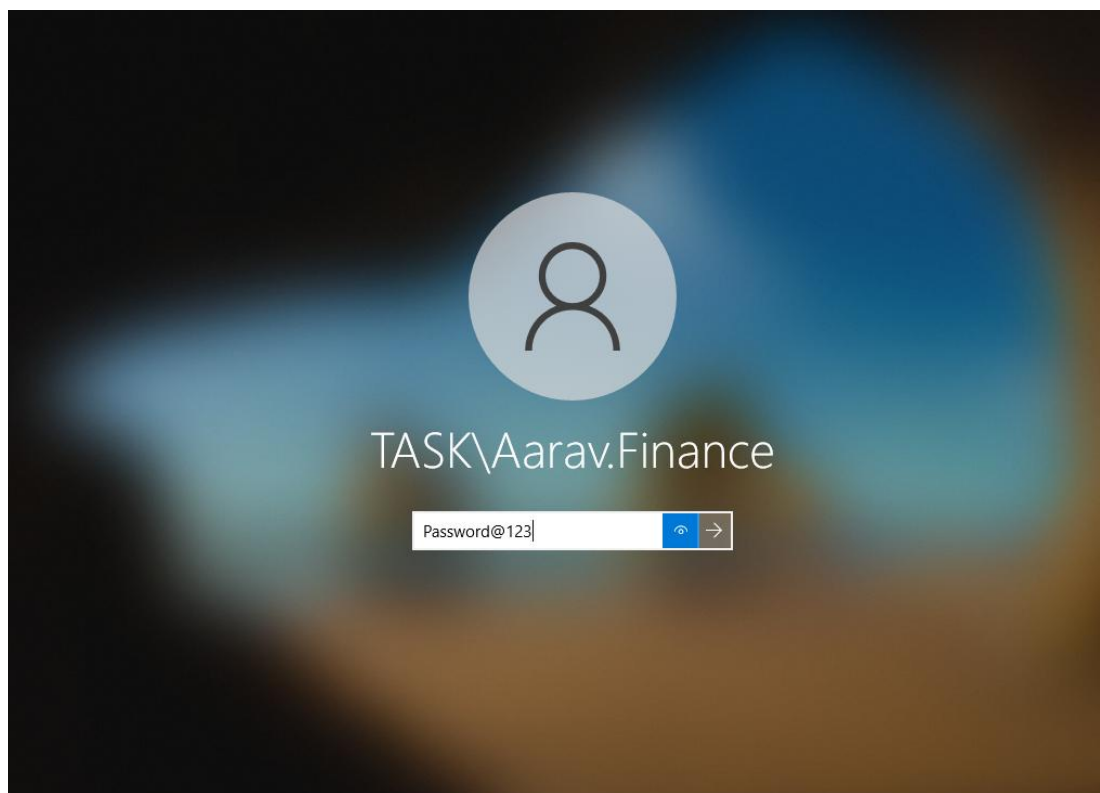
4.2 Reset Passwords for 5 Users and Enforce Change at Logon

Set-ADAccountPassword was used to assign a temporary password (Temp@1234) to 5 selected users. Set-ADUser -ChangePasswordAtLogon \$true was applied to enforce password reset at the next login. These changes were confirmed in ADUC under the Account tab.

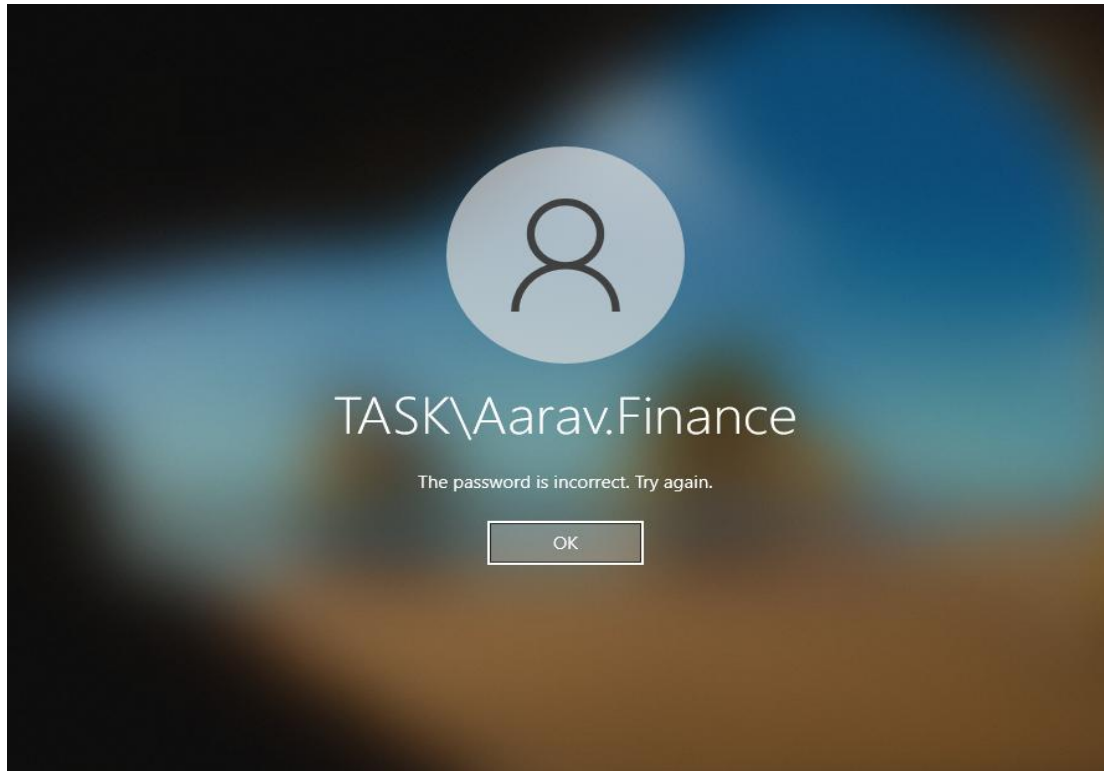
```
Select Administrator: Windows PowerShell
PS C:\Users\Administrator> Set-ADAccountPassword -Identity "Aarav.Finance" -Reset -NewPassword (ConvertTo-SecureString "NewPassword@123" -AsPlainText -Force)
PS C:\Users\Administrator> Set-ADUser -Identity "Aarav.Finance" -ChangePasswordAtLogon $true
PS C:\Users\Administrator>
PS C:\Users\Administrator> Set-ADAccountPassword -Identity "Diya.Finance" -Reset -NewPassword (ConvertTo-SecureString "NewPassword@123" -AsPlainText -Force)
PS C:\Users\Administrator> Set-ADUser -Identity "Diya.Finance" -ChangePasswordAtLogon $true
PS C:\Users\Administrator>
PS C:\Users\Administrator> Set-ADAccountPassword -Identity "Kabir.IT" -Reset -NewPassword (ConvertTo-SecureString "NewPassword@123" -AsPlainText -Force)
PS C:\Users\Administrator> Set-ADUser -Identity "Kabir.IT" -ChangePasswordAtLogon $true
PS C:\Users\Administrator>
PS C:\Users\Administrator> Set-ADAccountPassword -Identity "Rohan.HR" -Reset -NewPassword (ConvertTo-SecureString "NewPassword@123" -AsPlainText -Force)
PS C:\Users\Administrator> Set-ADUser -Identity "Rohan.HR" -ChangePasswordAtLogon $true
PS C:\Users\Administrator>
PS C:\Users\Administrator> Set-ADAccountPassword -Identity "Sneha.HR" -Reset -NewPassword (ConvertTo-SecureString "NewPassword@123" -AsPlainText -Force)
PS C:\Users\Administrator> Set-ADUser -Identity "Sneha.HR" -ChangePasswordAtLogon $true
PS C:\Users\Administrator>
```

(Reset passwords for 5 accounts and enforced password change at next login)

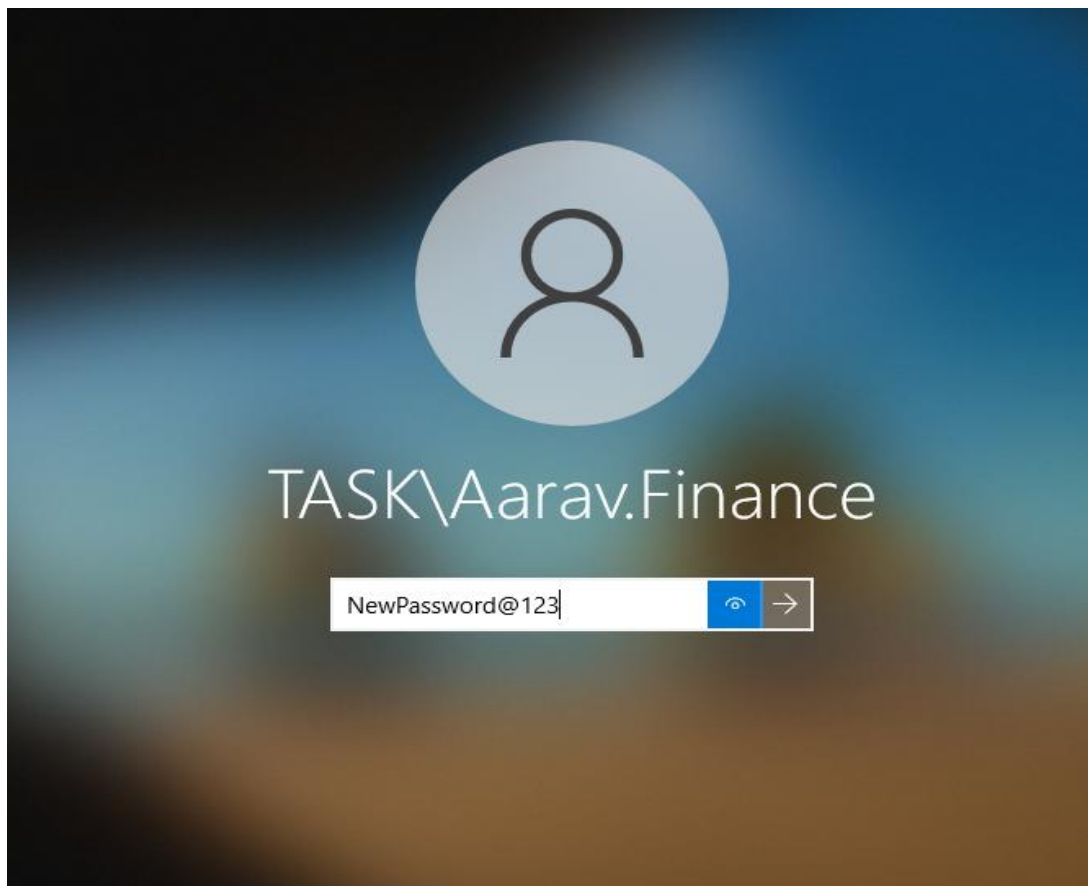
(NewPassword@123)



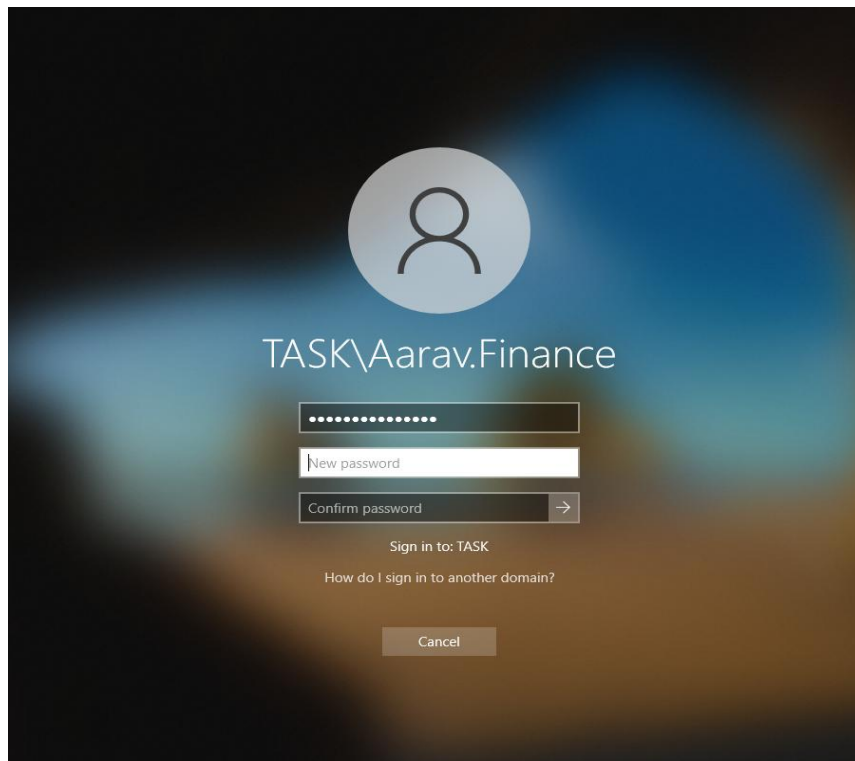
(tried to log in with old password)



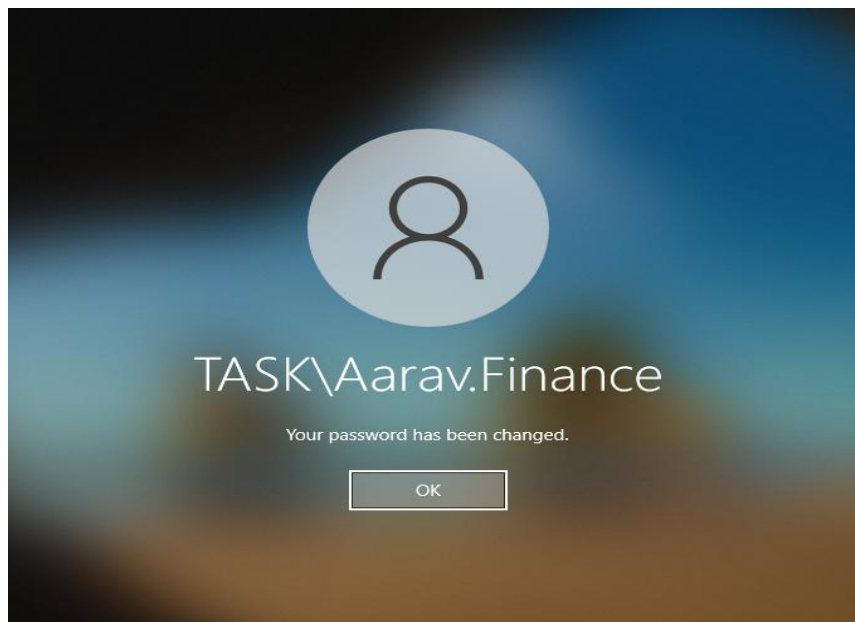
(can't login this means password is changed successfully)



(put new password and try to log in again)



(Now before logging in we have to put a new password)

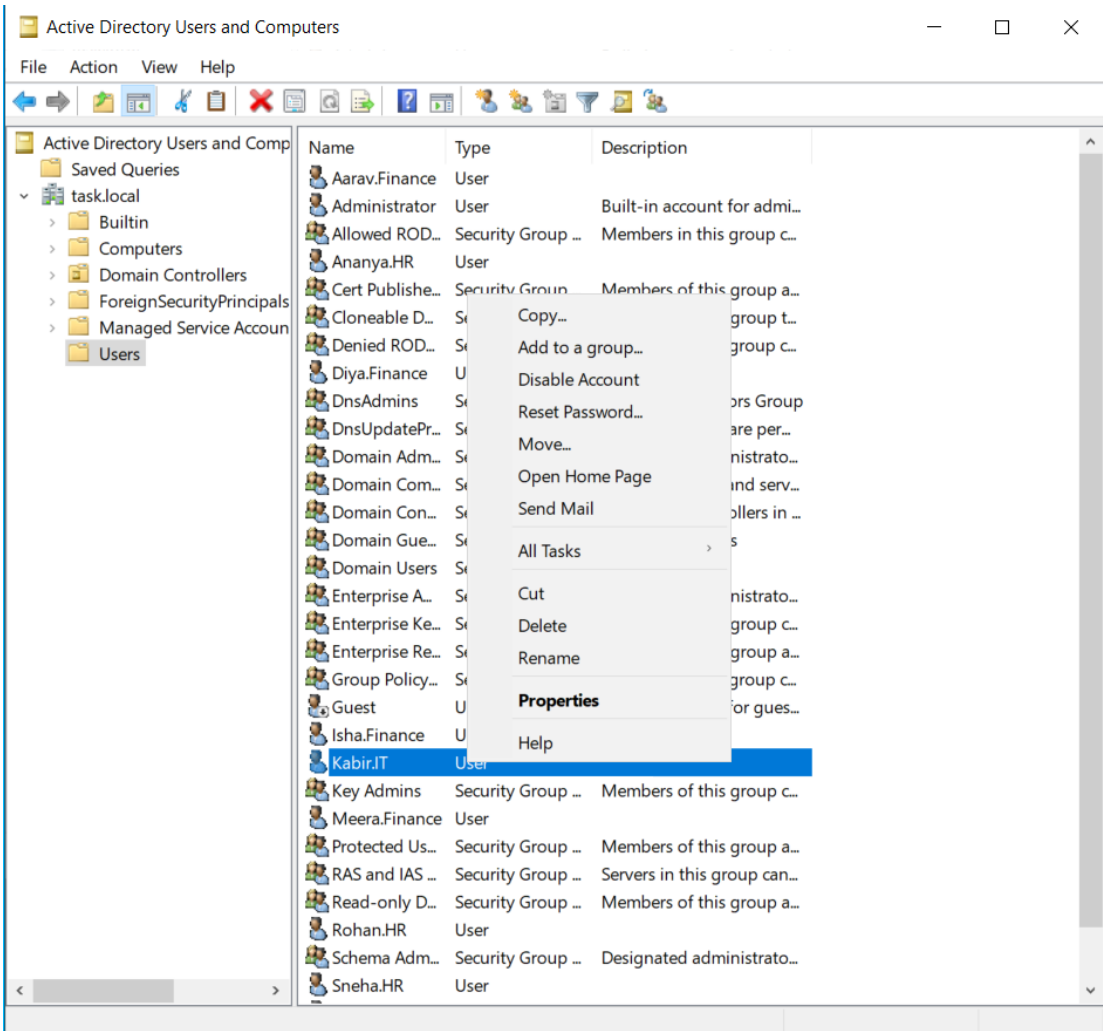


(Created a new password and logged in successfully)

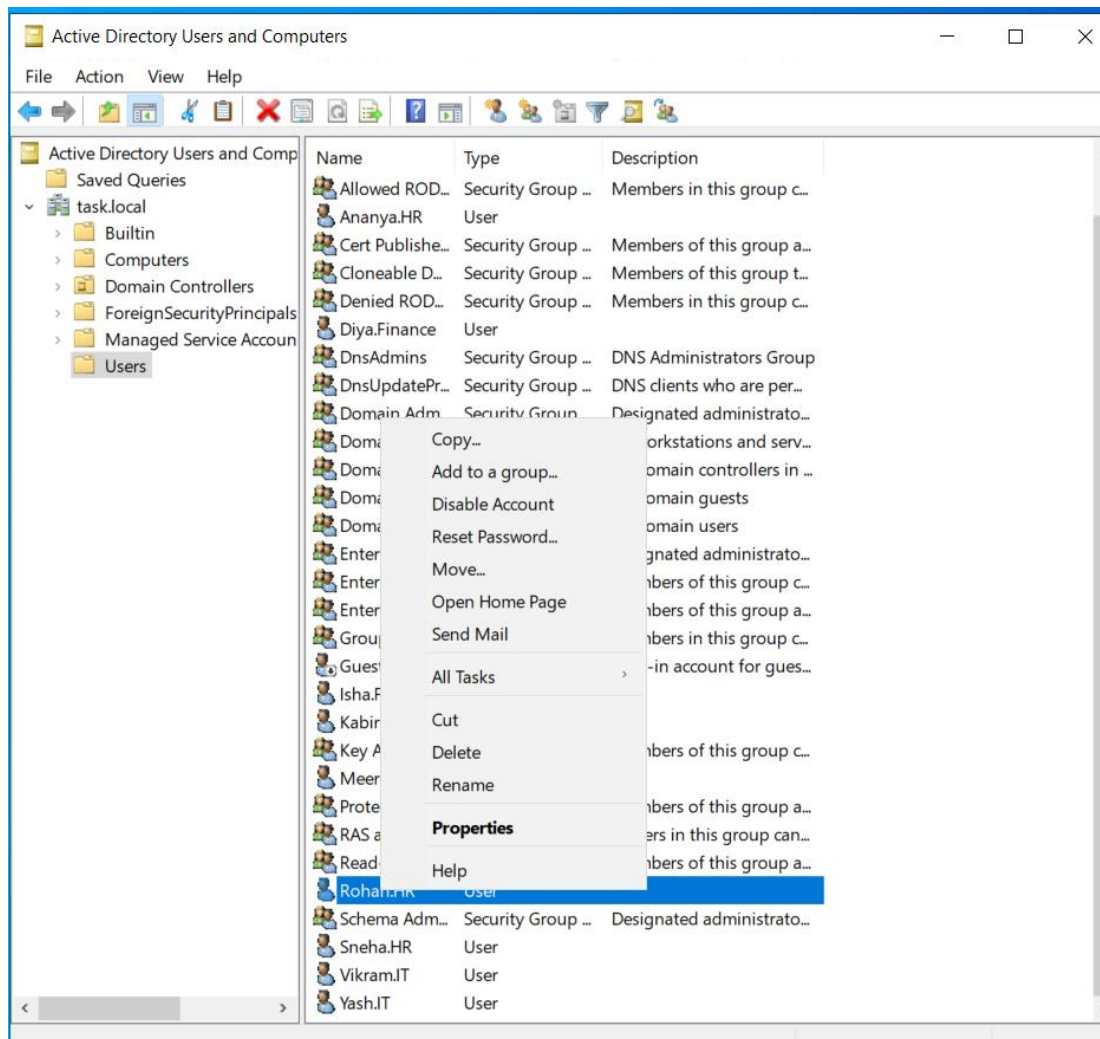
(Pass-Temp@1234)

4.3 Disable 2 Inactive Accounts

Disable-ADAccount was executed on two users marked for deactivation. Their status was confirmed by the presence of the downward arrow icon in ADUC.



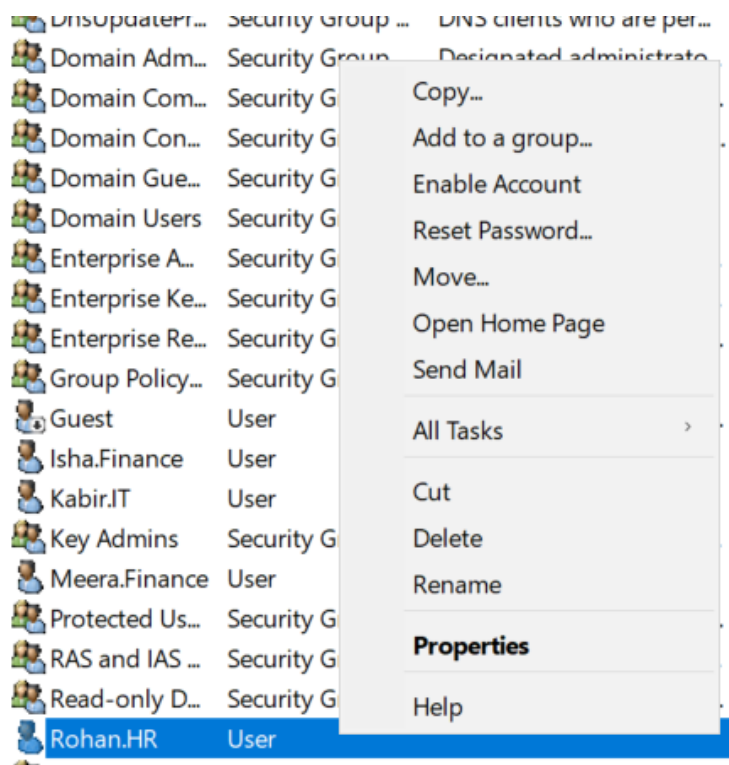
(Kabir.it-before)



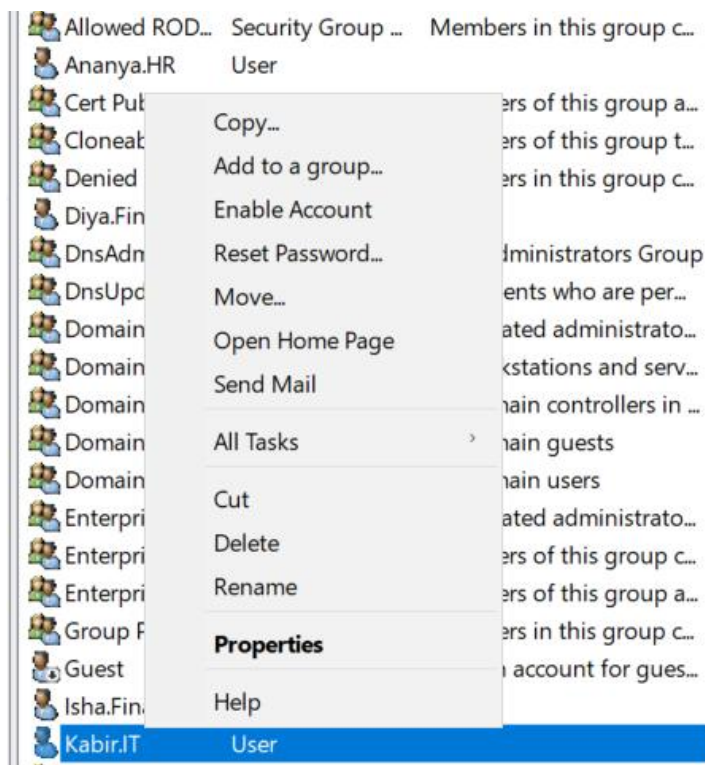
(Rohan.hr-before)

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Disable-ADAccount -Identity "Kabir.IT"
PS C:\Users\Administrator> Disable-ADAccount -Identity "Rohan.HR"
PS C:\Users\Administrator>
```

(PowerShell Commands: to disable accounts)



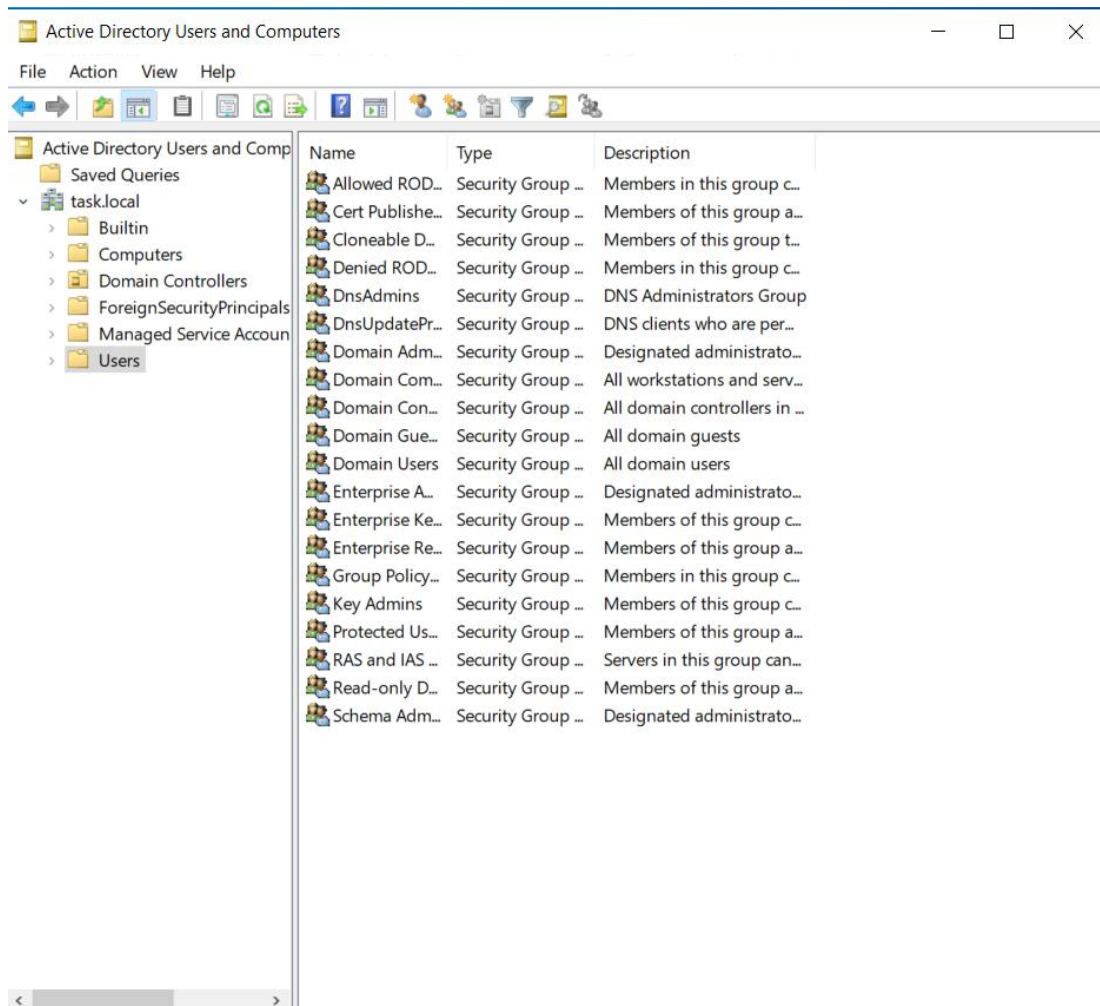
(Rohan .hr-after)



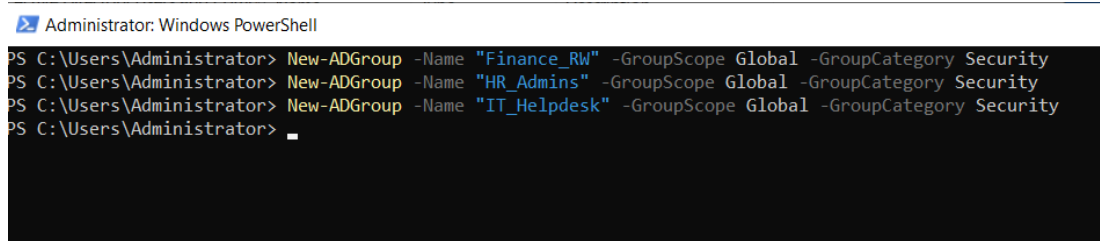
(Kabir.it – after)

4.4 Create 3 Security Groups

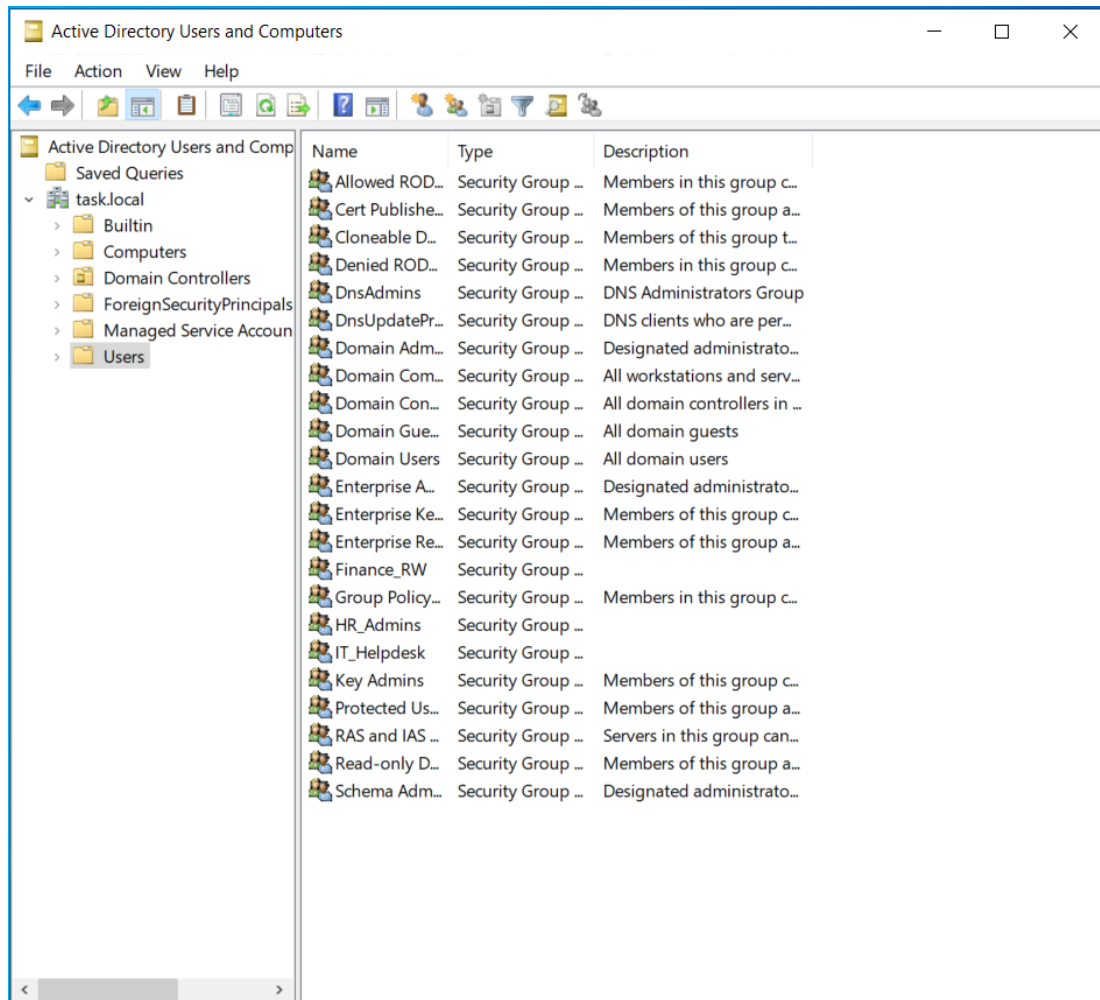
New-ADGroup was executed in PowerShell to create three global security groups: Finance_RW, HR_Admins, and IT_Helpdesk. These groups were confirmed in ADUC.



(No custom group exists)



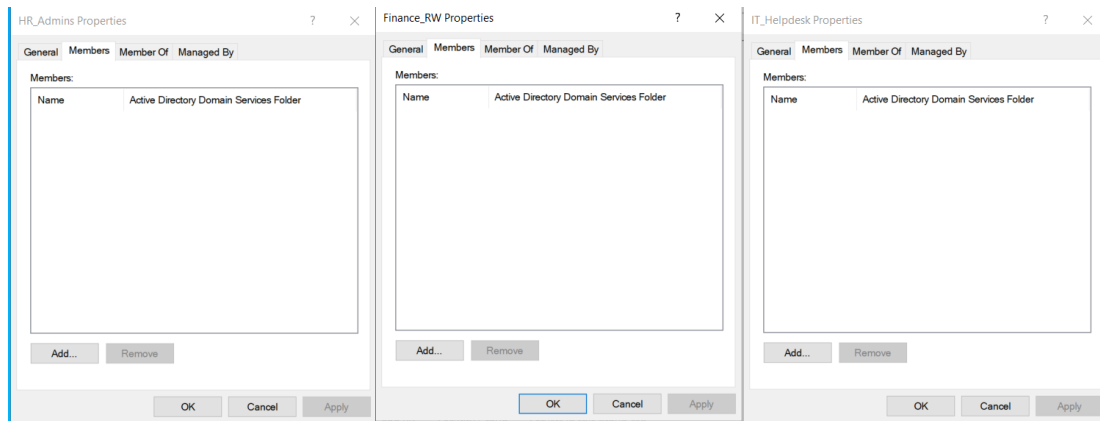
(Commands to create custom groups)



(Created 3 domain security groups for managing access and permissions by department)

4.5 Assign Users to Groups

Users were added to their respective groups using Add-ADGroupMember. Membership was confirmed through the Members tab in ADUC.



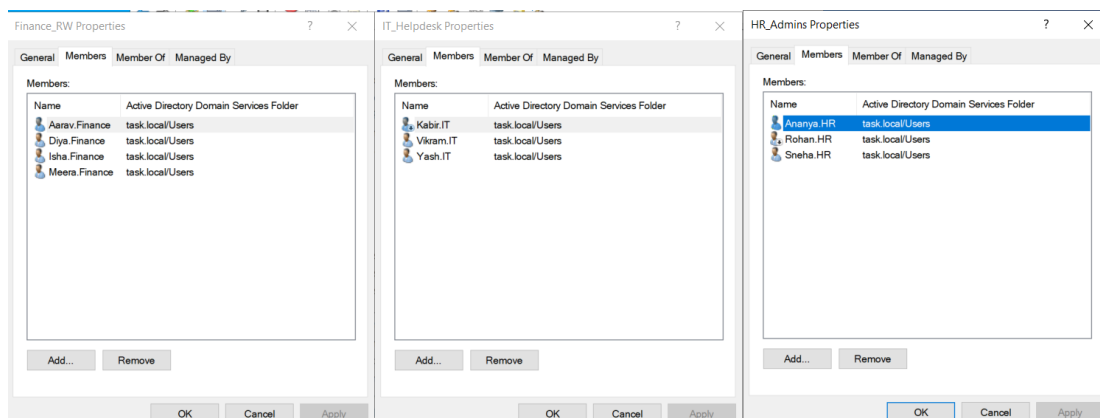
(Users are not assigned in security groups)

```

Administrator: Windows PowerShell
PS C:\Users\Administrator> Add-ADGroupMember -Identity "Finance_RW" -Members "Aarav.Finance","Diya.Finance","Meera.Finance","Isha.Finance"
>> Add-ADGroupMember -Identity "HR_Admins" -Members "Rohan.HR","Ananya.HR","Sneha.HR"
>> Add-ADGroupMember -Identity "IT_Helpdesk" -Members "Kabir.IT","Yash.IT","Vikram.IT"
PS C:\Users\Administrator>

```

(Powershell commands to assign users)



(The users are assigned)

4.6 Verify Group Membership

Get-ADGroupMember was used in PowerShell to confirm that each group contained the correct users.

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-ADGroupMember "Finance_RW"

distinguishedName : CN=Aarav.Finance,CN=Users,DC=task,DC=local
name              : Aarav.Finance
objectClass       : user
objectGUID        : 65df61d6-9571-4754-bdf8-ccdd6ff12e2b
SamAccountName    : Aarav.Finance
SID               : S-1-5-21-2559830969-548201595-822421515-1104

distinguishedName : CN=Diya.Finance,CN=Users,DC=task,DC=local
name              : Diya.Finance
objectClass       : user
objectGUID        : be853e9e-2fe7-4dd0-a2a8-2a257298f4db
SamAccountName    : Diya.Finance
SID               : S-1-5-21-2559830969-548201595-822421515-1105

distinguishedName : CN=Meera.Finance,CN=Users,DC=task,DC=local
name              : Meera.Finance
objectClass       : user
objectGUID        : dbc40001-679a-4a77-9016-e9676822863d
SamAccountName    : Meera.Finance
SID               : S-1-5-21-2559830969-548201595-822421515-1107

distinguishedName : CN=Isha.Finance,CN=Users,DC=task,DC=local
name              : Isha.Finance
objectClass       : user
objectGUID        : b628b134-85b7-4338-ac5d-41d013bf0224
SamAccountName    : Isha.Finance
SID               : S-1-5-21-2559830969-548201595-822421515-1111

PS C:\Users\Administrator>
```

(correct users had been added to finance group)

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-ADGroupMember "HR_Admins"

distinguishedName : CN=Rohan.HR,CN=Users,DC=task,DC=local
name              : Rohan.HR
objectClass       : user
objectGUID        : 1a587bd1-64d2-4439-9e5b-46188c416f1c
SamAccountName    : Rohan.HR
SID               : S-1-5-21-2559830969-548201595-822421515-1108

distinguishedName : CN=Ananya.HR,CN=Users,DC=task,DC=local
name              : Ananya.HR
objectClass       : user
objectGUID        : b8bba831-2dd6-4682-af2e-452313cb3a1d
SamAccountName    : Ananya.HR
SID               : S-1-5-21-2559830969-548201595-822421515-1109

distinguishedName : CN=Sneha.HR,CN=Users,DC=task,DC=local
name              : Sneha.HR
objectClass       : user
objectGUID        : 1f40ef65-963b-4a9a-b854-618a69004bcb
SamAccountName    : Sneha.HR
SID               : S-1-5-21-2559830969-548201595-822421515-1113

PS C:\Users\Administrator> _
```

(correct users had been added to hr group)

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-ADGroupMember "IT_Helpdesk"

distinguishedName : CN=Kabir.IT,CN=Users,DC=task,DC=local
name              : Kabir.IT
objectClass       : user
objectGUID        : 9be77118-e3a0-4ea8-b660-d79fb3f124c3
SamAccountName    : Kabir.IT
SID               : S-1-5-21-2559830969-548201595-822421515-1106

distinguishedName : CN=Yash.IT,CN=Users,DC=task,DC=local
name              : Yash.IT
objectClass       : user
objectGUID        : 4327f7f9-6734-4998-8d36-a9c46c025b0e
SamAccountName    : Yash.IT
SID               : S-1-5-21-2559830969-548201595-822421515-1110

distinguishedName : CN=Vikram.IT,CN=Users,DC=task,DC=local
name              : Vikram.IT
objectClass       : user
objectGUID        : 72b1782b-7e4c-46c6-8ad7-8b9cb2fdf877
SamAccountName    : Vikram.IT
SID               : S-1-5-21-2559830969-548201595-822421515-1112

PS C:\Users\Administrator>
```

(correct users had been added to group)

5. Recommendations

- Implement a naming convention policy for all future user and group accounts to ensure clarity and consistency across departments.
- Periodically review and disable inactive accounts to minimize security risks associated with unused credentials.
- Enforce password expiration policies domain-wide and ensure that service accounts are excluded with justification and monitoring.
- Maintain documentation of all group memberships to support access control reviews and compliance audits.
- Automate repetitive user and group tasks using PowerShell scripts or task scheduling to reduce administrative overhead.
- Regularly audit group membership to prevent privilege creep and ensure users retain only the access necessary for their roles.

6. Conclusion

The successful execution of foundational Active Directory tasks including user creation, password enforcement, account deactivation, group configuration, and membership validation has established a secure and organized identity framework within the domain. By aligning users with their respective departments and assigning them to logically structured security groups, we have laid the groundwork for role-based access control and streamlined policy management. All operations were performed using a combination of PowerShell and graphical tools to ensure both efficiency and visibility. This approach not only strengthens administrative control but also prepares the environment for future security and compliance configurations such as group policy enforcement and auditing.